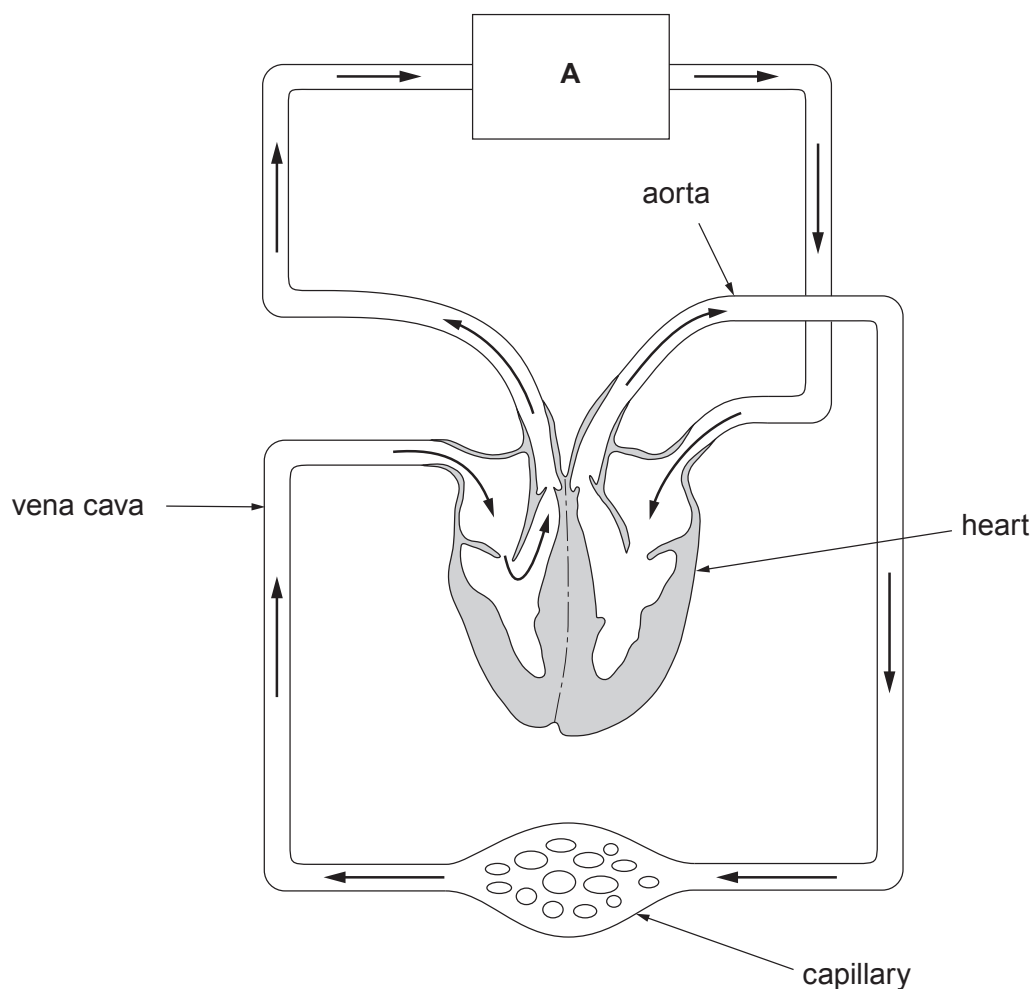


Answer **all** questions.

1. (a) **Image 1.1** shows part of the human circulatory system.

Image 1.1



From **Image 1.1**:

- (i) Underline the name of organ **A** from the choices below. [1]

brain lung kidney stomach

- (ii) Use the **blood vessels named in Image 1.1**, to state the blood vessel that:

I. is an artery; [1]

II. releases oxygen and glucose to the body cells. [1]

.....

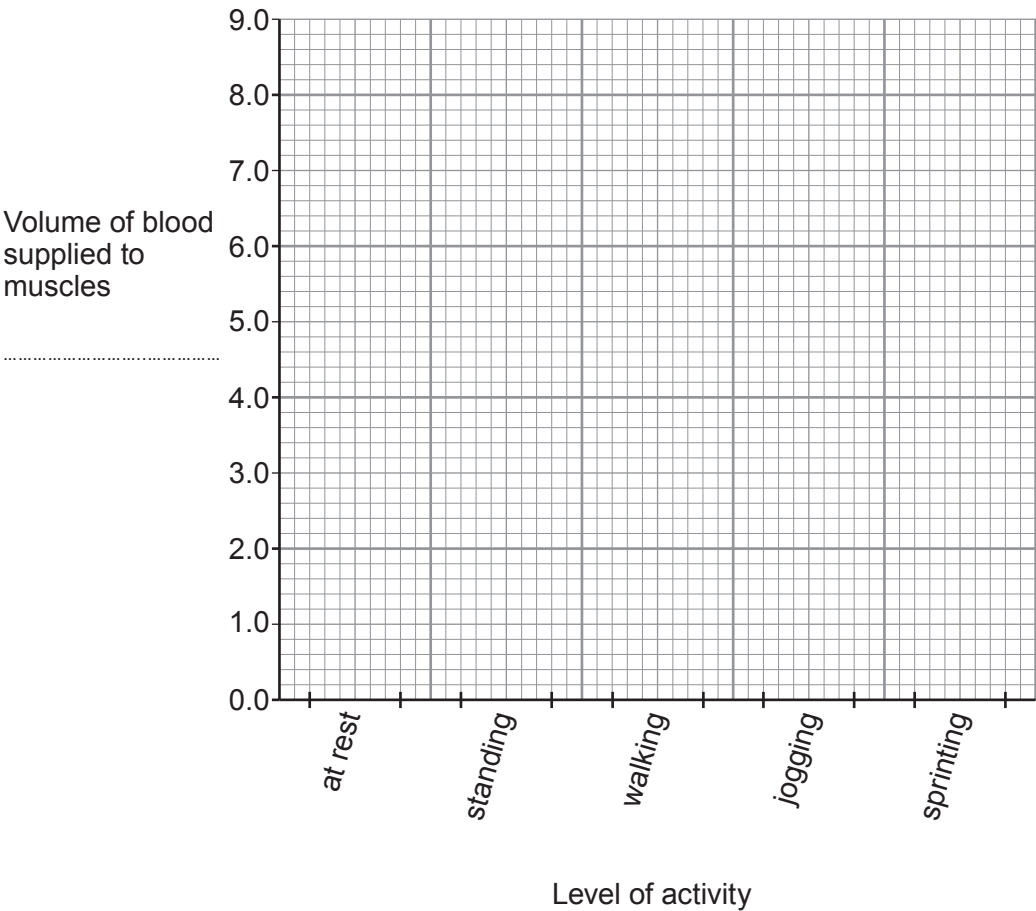


(b) **Table 1.2** shows the volume of blood supplied to the muscles of an athlete at increasing levels of activity.

Table 1.2

Level of activity	Volume of blood supplied to muscles (dm ³ /min)
at rest	2.0
standing	2.4
walking	3.6
jogging	5.0
sprinting	8.0

- (i) Draw a **bar chart** of the data shown in **Table 1.2** on the grid below by: [3]
- I. completing the label for the vertical axis by **adding the units**;
 - II. using a **ruler** to draw the bars.



- (ii) Calculate the difference in blood volume supplied per minute to the muscles when the level of activity changes from walking to sprinting. [1]

Difference = dm³/min

- (iii) I. State how increasing the level of activity changes the volume of blood supplied to muscles. [1]

.....

- II. Complete the sentence below by underlining the correct term in each bracket to explain how this change helps muscles work harder. [2]

This change provides more (**carbon dioxide / oxygen / nitrogen**)
for (**respiration / photosynthesis / digestion**).

